Build a complete prediction

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Connect net force, velocity change, and final position in one model.

SUCCESS CRITERIA - I can...

1. Connect force, acceleration and position consistently.
2. Show quantities and units at every stage.
3. Explain the assumptions and limits of the prediction.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

Maximum static friction is 50 N. A 20 N push leaves a box still. Find actual friction. A 60 N push starts sliding with 40 N kinetic friction; find net force.

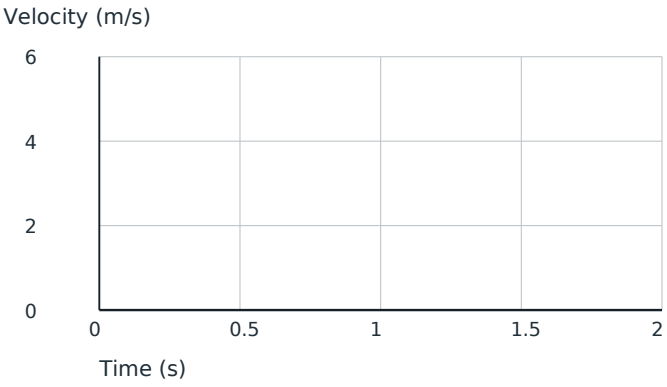
Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $a = \Sigma F/m$; $v_f = v_i + a\,\Delta t$; $x_f = x_i + \frac{1}{2}(v_i + v_f)\Delta t$. Constant acceleration; use one frame. i = initial; f = final. Apply each direction separately.

Worked reasoning: A 2 kg cart starts from rest under 6 N right and 2 N left for 2 s. Net force 4 N gives $a=2\ m/s^2$. Its displacement is $\frac{1}{2}\times2\times2^2=4$ m right.

Your turn: Draw the force diagram and the matching velocity-time triangle. Check the 4 m using its area.



Marzano check - circle: ① ② ③

03 Start with forces

A 2 kg cart starts at rest. There are 6 N right and 2 N left. Find net force and acceleration.

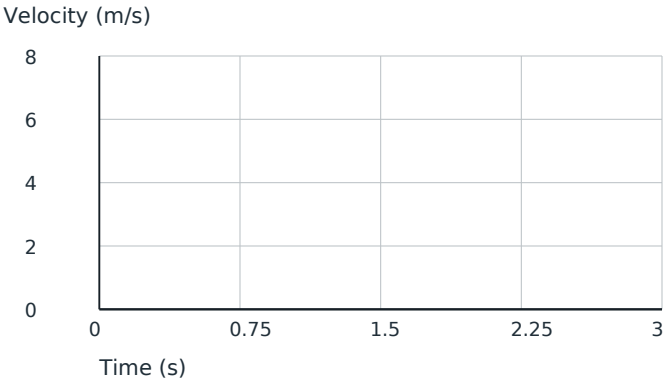


The dot represents the object. Mass 2 kg; starts at rest

Marzano check - circle: ① ② ③

04 Predict motion

The forces on the cart in question 3 stay constant for 3 s. Draw its velocity-time graph, then find final velocity, average velocity and displacement. If the cart starts at the 1 m mark, where does it finish?



Marzano check - circle: ① ② ③

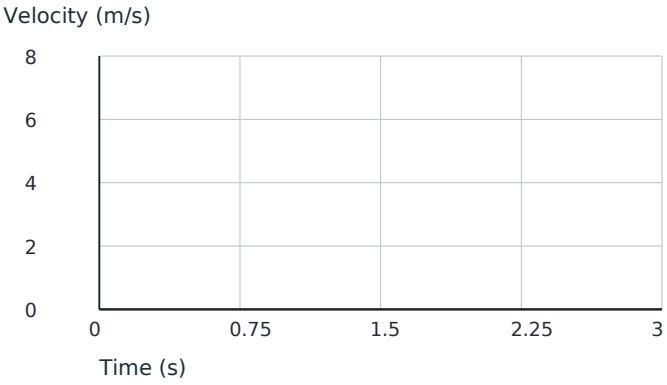
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Continue down each column.

05 Explain a limit

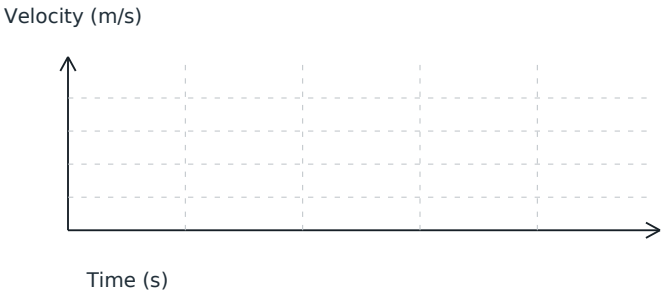
Your prediction in question 4 assumes constant net force. Name a change that could break that assumption. Draw how the velocity-time line would differ if a larger, constant opposing friction force acted throughout.



Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

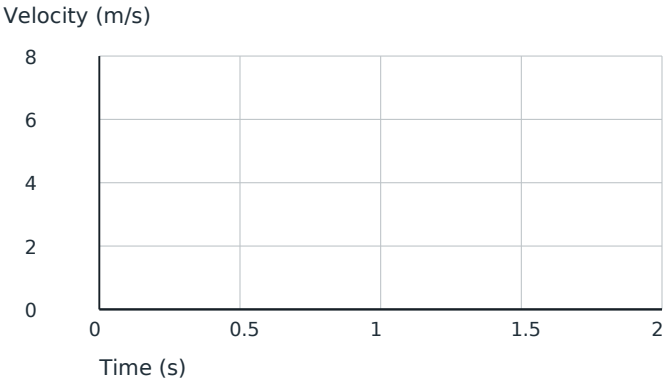
A cart accelerates uniformly from rest to 14 m/s east in 5 s. Find acceleration, average velocity and displacement. Sketch the velocity-time model; label the given values.



Marzano check - circle: ① ② ③

07 Exit check

A 3 kg rover starts from rest with a constant 9 N net force to the right for 2 s. Find acceleration and final velocity. Draw a graph that checks your result, and state one assumption.



Marzano check - circle: ① ② ③

08 STRETCH: test the model

Return to the worked example in problem 2. A second cart must cover the same 4 m in 2 s from rest but has twice the mass. What net force is required? Show your chain of reasoning.



Add and label force arrows.

The second cart has twice the worked example's mass.

Marzano check - circle: ① ② ③